SYSC 5108 Exam Study: Assignment 3, Question 1

Entropy of a Bernoulli random variable and its maximum

Source question: `SYSC 5108 W26 Assignment 3.pdf`, Question 1

Question

Maximum entropy is often used to select an a priori distribution when there is very little information about the random variable. Consider the Bernoulli distribution

$$p(x) = \begin{cases} p, & x = 1, \\ 1 - p, & x = 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find its entropy $H(X)$ and prove analytically that $H(X)$ achieves its maximum when

$$p = \frac{1}{2}.$$

Step 1: Recall the entropy definition

Chapter 3 slide 47 defines entropy as the expected information carried by a random variable:

$$H(X) = -\sum_x p(x) \log p(x).$$

Goodfellow, Bengio, and Courville define Shannon entropy the same way in Chapter 3:

$$H(X) = \mathbb{E}_{X\sim P}[I(X)] = -\mathbb{E}_{X\sim P}[\log P(X)].$$

Unless otherwise specified, the course and textbook use natural logarithms, so the units are nats. If base-2 logarithms are used instead, the units are bits.

The slide convention

$$0 \log 0 = 0$$

handles impossible outcomes.

Step 2: Apply the definition to a Bernoulli variable

A Bernoulli random variable has two possible outcomes:

$$P(X = 1) = p, \qquad P(X = 0) = 1 - p.$$

So the entropy is

$$H(X) = -\left[P(X = 1) \log P(X = 1) + P(X = 0) \log P(X = 0)\right].$$

Substitute the probabilities:

$$\boxed{H(X) = -p \log p - (1 - p) \log(1 - p)}.$$

This can also be written as

$$H(X) = (p - 1) \log(1 - p) - p \log p.$$

Step 3: Find the critical point

Differentiate $H(p)$ with respect to p:

$$H(p) = -p \log p - (1-p)\log(1-p).$$

For the first term,

$$\frac{d}{dp}[-p \log p] = -(\log p + 1).$$

For the second term,

$$\frac{d}{dp}[-(1-p)\log(1-p)] = \log(1-p) + 1.$$

Therefore

$$\frac{dH}{dp} = -(\log p + 1) + \log(1-p) + 1.$$

Simplify:

$$\frac{dH}{dp} = \log(1-p) - \log p = \log\left(\frac{1-p}{p}\right).$$

Set the derivative equal to zero:

$$\log\left(\frac{1-p}{p}\right) = 0.$$

Exponentiate both sides:

$$\frac{1-p}{p} = 1.$$

So

$$1 - p = p.$$

Hence

$$\boxed{p = \frac{1}{2}.}$$

Step 4: Prove the critical point is a maximum

Compute the second derivative:

$$\frac{dH}{dp} = \log(1-p) - \log p.$$

Differentiate again:

$$\frac{d^2H}{dp^2} = -\frac{1}{1-p} - \frac{1}{p}.$$

Combine terms:

$$\frac{d^2H}{dp^2} = -\left(\frac{1}{1-p} + \frac{1}{p}\right).$$

For $0 < p < 1$,

$$\frac{1}{1-p} > 0, \qquad \frac{1}{p} > 0,$$

so

$$\frac{d^2H}{dp^2} < 0.$$

Therefore $H(p)$ is strictly concave on $(0, 1)$, and the only critical point is the global maximum.

At $p = \frac{1}{2}$:

$$H\left(\frac{1}{2}\right) = -\frac{1}{2}\log\frac{1}{2} - \frac{1}{2}\log\frac{1}{2}.$$

Since

$$\log\frac{1}{2} = -\log 2,$$

we get

$$H\left(\frac{1}{2}\right) = \log 2.$$

Thus the maximum entropy is

$$\boxed{H_{\max} = \log 2 \text{ nats.}}$$

If using base-2 logarithms, this is

$$\boxed{H_{\max} = 1 \text{ bit.}}$$

Step 5: Check the boundary cases

At $p = 0$:

$$H(0) = -0\log 0 - (1)\log(1) = 0.$$

At $p = 1$:

$$H(1) = -(1)\log(1) - 0\log 0 = 0.$$

These are deterministic cases: if $p = 0$, X is always 0; if $p = 1$, X is always 1. There is no uncertainty, so entropy is zero.

At $p = \frac{1}{2}$, both outcomes are equally likely:

$$P(X = 0) = P(X = 1) = \frac{1}{2}.$$

This is the most uncertain Bernoulli distribution, so it has the maximum entropy.

Step 6: Exam Interpretation

This result explains the maximum-entropy principle in the Bernoulli case. If there is no prior information favoring $X = 0$ over $X = 1$, choose the uniform Bernoulli distribution:

$$P(X = 0) = P(X = 1) = \frac{1}{2}.$$

This is the least committed prior over the two outcomes because it has the largest entropy.

A concise exam answer could be:

> For a Bernoulli random variable,
>
> $$H(X) = -\sum_x p(x)\log p(x) = -p\log p - (1-p)\log(1-p).$$

Differentiate:
$$\frac{dH}{dp} = \log(1-p) - \log p = \log\left(\frac{1-p}{p}\right).$$
Setting this to zero gives $(1-p)/p = 1$, so $p = 1/2$. The second derivative is

$$\frac{d^2H}{dp^2} = -\frac{1}{1-p} - \frac{1}{p} < 0$$

for $0 < p < 1$, so this critical point is a maximum. The maximum value is

$$H(1/2) = \log 2$$

nats, or 1 bit if base-2 logarithms are used.

Cross References Used

- Assignment: `SYSC 5108 W26 Assignment 3.pdf`, Question 1.
- Local submitted Assignment 3 PDF checked for comparison, Question 1 response.
- Slides: Chapter 3 slide deck, slide 47 on entropy as average information and the convention $0 \log 0 = 0$.
- Textbook: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 3, Section 3.9.1 on the Bernoulli distribution.
- Textbook: Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 3, Section 3.13 on information theory and Shannon entropy.